Rational Functions and Domain

A **rational function** is a quotient of polynomials:

$$f(x) = \frac{P(x)}{Q(x)}, \qquad Q(x) \neq 0.$$

Its domain includes every real number except the real zeros of the *original* denominator.

1. Factor the denominator and find its real zeros.
2. Record the excluded values before canceling common factors.
3. Simplify when possible, while keeping the original exclusions.

If a common factor cancels and the reduced denominator is nonzero there, the graph has a **hole**. Use the reduced formula to find its height.

- $f(x) = \frac{x^2-1}{x-1}$, $x \neq 1$.

Find the Domain

Find the domain of

$$f(x) = \frac{x+4}{x^2+x-6}.$$

Show the factorization and write the domain in interval notation.

In order to find the domain, set $x^2+x-6=0$

use "ac skill". So $ac=-6$, the number we find is 3 and -2.

So $x^2+x-6=0$

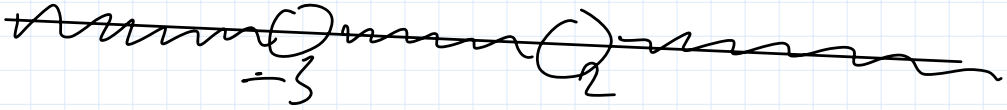
$x^2+3x-2x-6=0$

$-2x-6+x^2+3x$

$-2(x+3)+x(x+3)=0$

$(x+3)\cdot(x-2)=0$

So $x=-3$, $x=2$ is not in the domain.



$(-\infty,-3)\cup(-3,2)\cup(2,\infty)$

Special Domain Cases

Find the domain of each function. Explain any excluded values.

(a) $g(x) = \dfrac{2x-5}{4}$ (b) $h(x) = \dfrac{3}{x^2+9}$.

(a) Domain $= (-\infty, \infty)$

x^2+9 is never 0.

(b) Domain $(-\infty, \infty)$

$a^2-b^2=(a+b)(a-b)$

A Canceled Factor and a Hole

Let

$$f(x)=\frac{x^2-9}{x-3}.$$

Find the domain, simplify the formula, locate the hole, and sketch the graph.

- $x\neq 3$ is domain
- $f(x)=\frac{x^2-9}{x-3}$

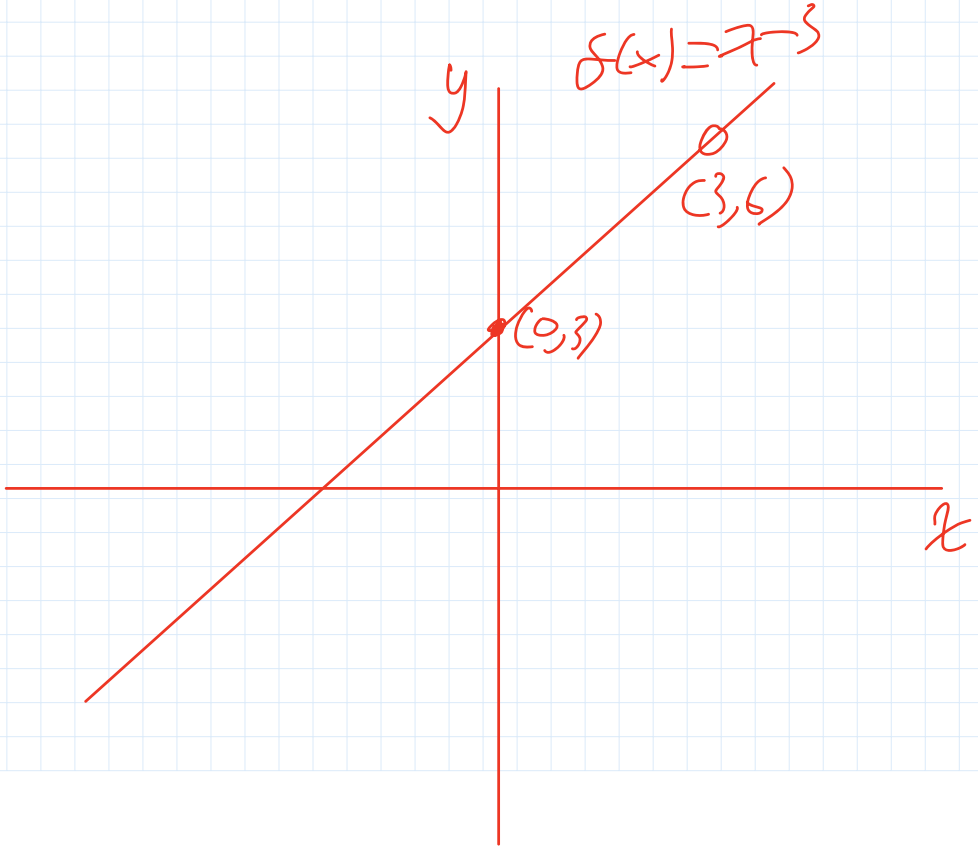
$=\frac{(x+3)(x-3)}{x-3}$

$=x+3$ ← $f(3)=3+3=6$

$f(x)=x+3$, with domain $x\neq 3$.

$(3,6)$ is the hole.

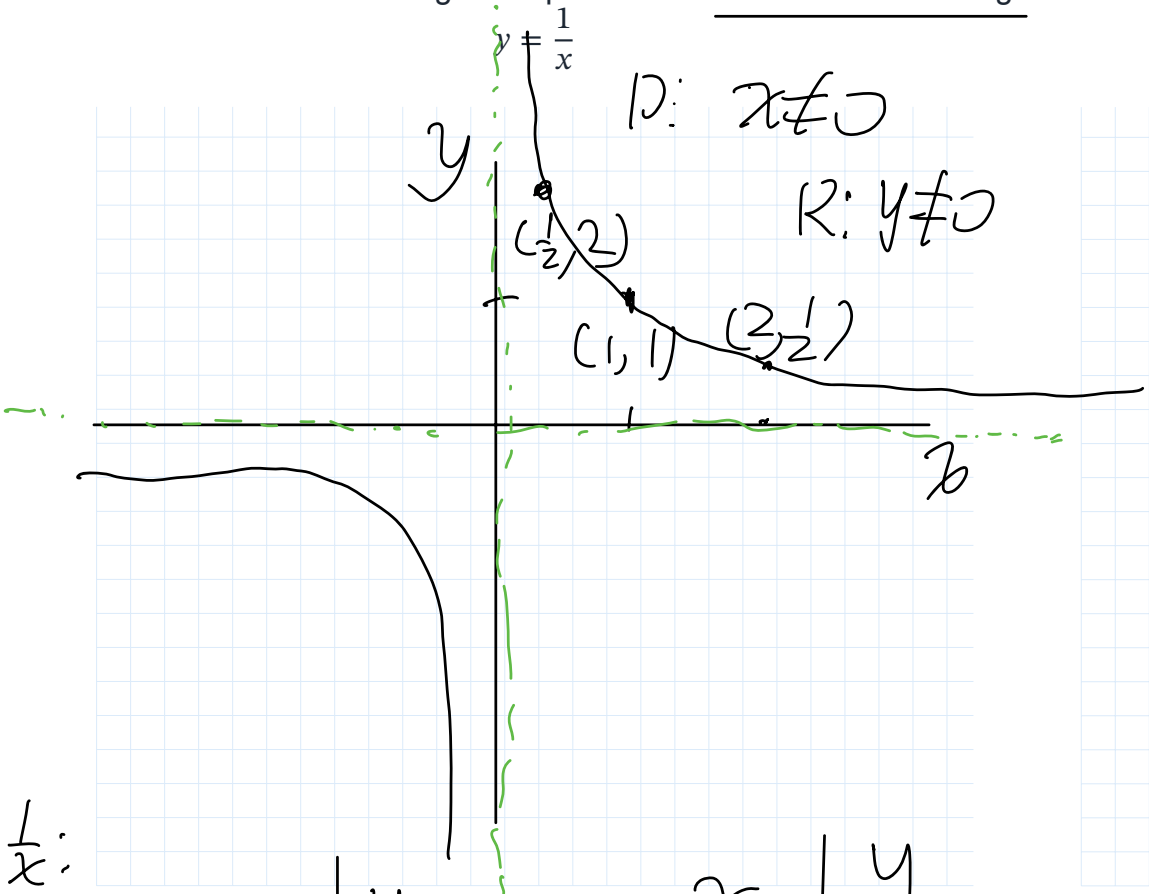
$f(3)=\frac{3^2-9}{3-3}=\frac{0}{0}$?

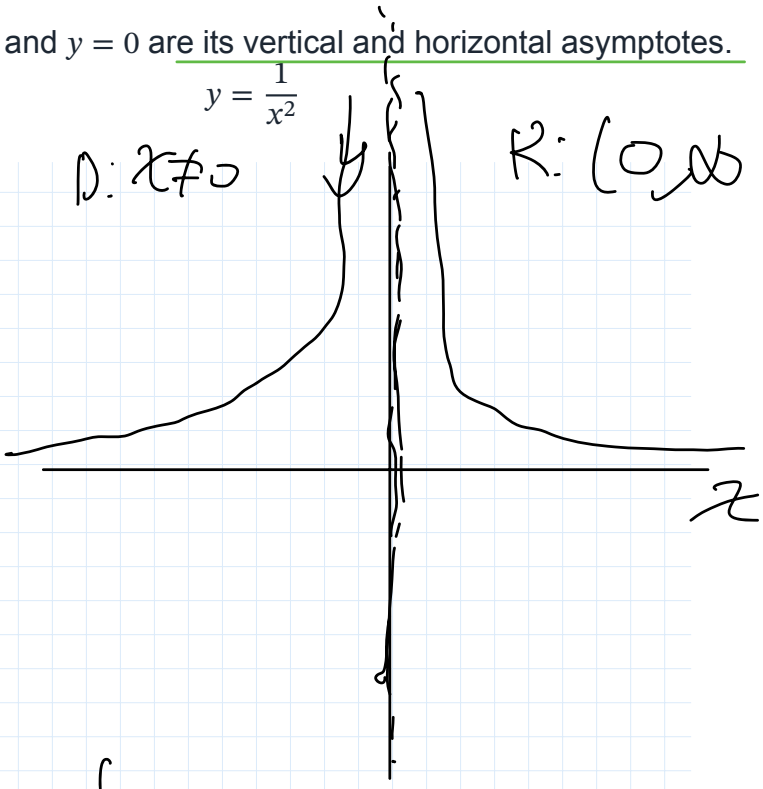


- the graph of $\frac{x^2-9}{x-3}$ is same as $x+3$.

Two Elementary Reciprocal Functions

Sketch each function using a few points. State its domain and range. The lines $x = 0$ and $y = 0$ are its vertical and horizontal asymptotes.





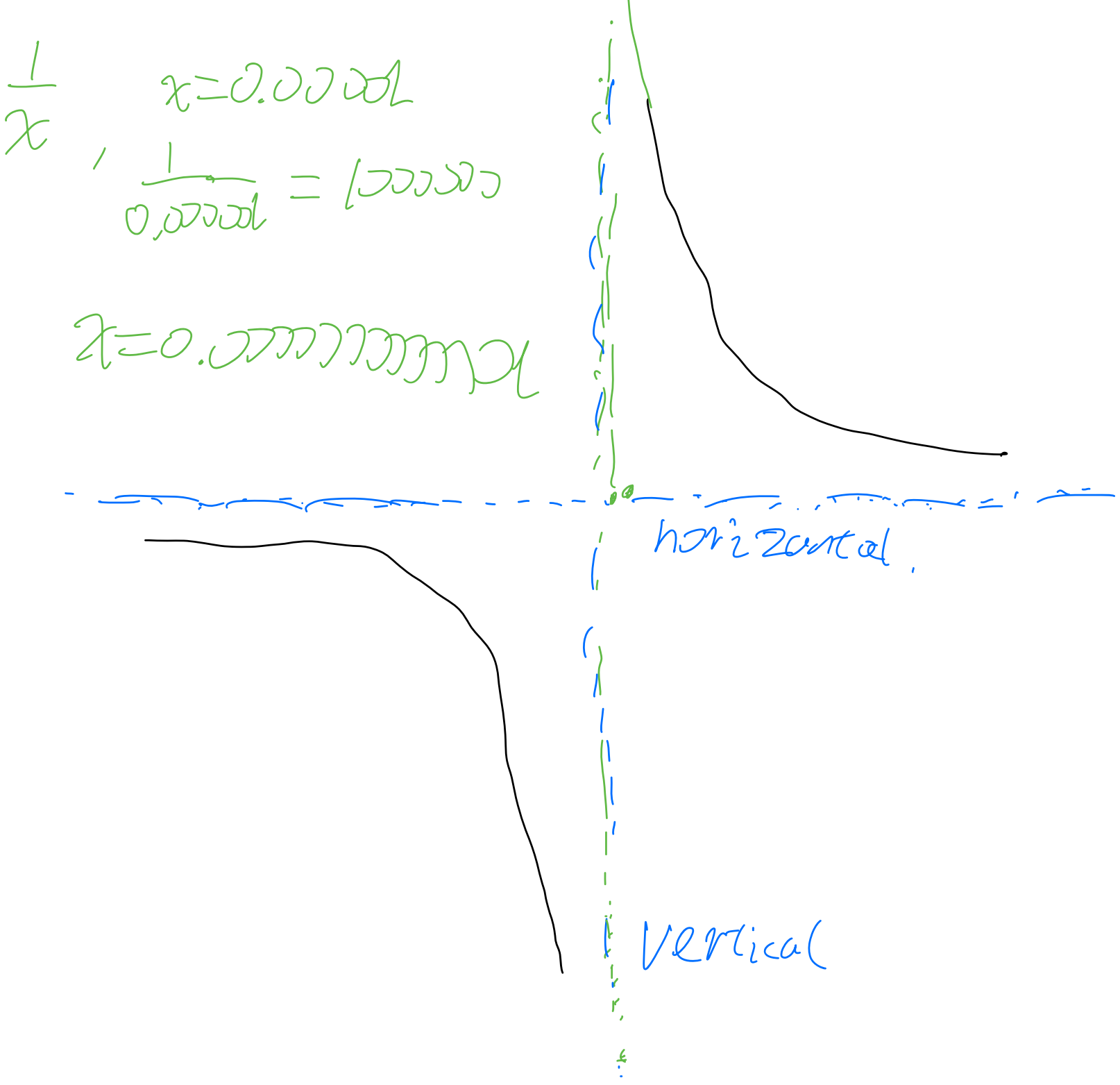
$\frac{1}{x}$:

x	y
-2	$\frac{1}{-2}$
-1	-1
1	1
2	$\frac{1}{2}$

x	y
$-\frac{1}{2}$	-2
$\frac{1}{2}$	2

$\frac{1}{x^2}$

x	y
1	$\frac{1}{1^2} = 1$
2	$\frac{1}{2^2} = \frac{1}{4}$
-1	$\frac{1}{(-1)^2} = 1$
-2	$\frac{1}{4}$


horizontal
Vertical

Transform a Reciprocal Function

Use the elementary function $y = \frac{1}{x}$ to sketch

$$g(x) = -\frac{2}{x-1} + 3. \quad = -2 \cdot \frac{1}{x-1} + 3$$

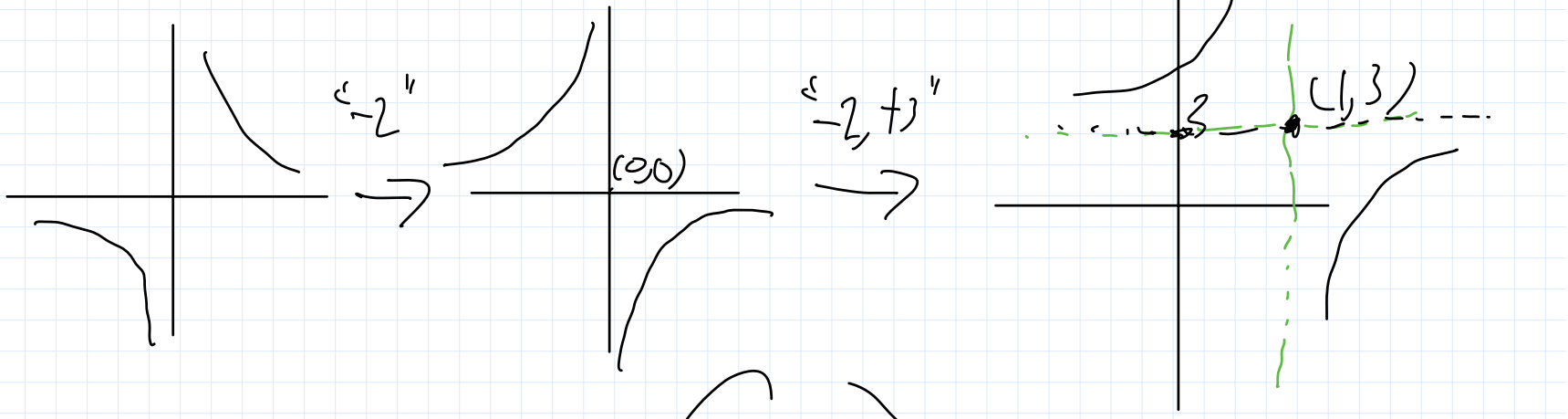
Describe the transformations. State the domain, range, and both intercepts.

transformation: -2 says Reflect by x-axis. Stretch vertically by 2.

$-2, +3$ says: right by 1, up by 3.

D: $x \neq 1$

R: $y \neq 3$



y-int: $g(0) = -\frac{2}{0-1} + 3 = 5$. $(0,5)$

x-int: $-\frac{2}{x-1} + 3 = 0$ $\quad \frac{-2}{x-1} = -3$ $\quad -2 = -3(x-1)$ $\quad x = \frac{5}{3}$. $\left(\frac{5}{3}, 0\right)$

Intercepts and Excluded Inputs

For

$$r(x) = \frac{x(x-2)}{x+1},$$

find the domain and all x- and y-intercepts.

D: $x \neq 2$

- x-int: $x(x-2)=0$

$x=0$ or $x=2$

$(0, r(0))$, $(2, r(2))$ $\quad r(2) = 2(2-0)=0$

$(0,0)$, $(2,0)$

- y-int: $r(0) = \frac{0(0-2)}{0+1} = 0$

$(0,0)$